Lesson 4

Equipment: Raspberry Pi Python code editor, provided sheets, pens/pencils

Summary

- This lesson is themed around decision-making and introduces the if/else construct in Python.

Learning Intentions

- I can consider the thought process behind making decisions in my own life
- I can model a decision-making process using an if/else structure within a Python program
- I can recognise real-world uses of if/else logic in modern applications

Curriculum Integration

- TCH 2-13a: pupils should be able to correctly structure a section of code using if/else statements. Pupils should also be able to navigate other sections of code that use if/else statements and understand which branch is selected given different code conditions.
- TCH 2-14a: pupils should be able to explain the operation and function of if/else statements in Python.
- TCH 2-15a: pupils should be able to implement their own programs using an if/else statement in Python.

Plan

Starter

- Prompt for pupils: Can computers make decisions? If so, what kinds of decisions do computers make? Or, is it really a human behind every decision?

Unplugged – Coats Outside (20 minutes)

- Pupils are asked to decide if they are planning on wearing a coat when going outside later.
- They are prompted to work in teams to come up with 3 factors that might influence their decision.
- As a class, work on writing rules to model this decision-making. E.g. statements such as “if it is sunny, I will not wear a coat”. Encourage the use of if/else language here.
- Pupils should then work in their teams to come up with and write down an if/else rule for one of the factors they came up with.
- Ex. “if it is rainy outside, I will wear a coat. Else, I will not wear a coat.”

Taught section (10 minutes)

- Pupils see the if/else construct in Python. Show pupils the examples on slides 16-17. VPC can be used to teach this section by asking pupils to predict what will be printed out in each case. In order to answer these questions correctly, pupils will need to have an understanding of if/else branching.

Coding Activity (30 minutes)

- Pupils should incorporate an if/else statement into their conversation programs. They are encouraged to introduce the variable as a way to read in user input. For example, pupils may ask what a person's favourite colour is. If the user responds with their own favourite colour, they might give a response stating this. Otherwise, they may print a more generic response. Pupils who struggle to do this could be encouraged to create a program similar to the final examples on slide 17. Creating a "guessing game" for a partner is a suitable alternative for pupils who struggle to incorporate a variable into their conversation program.

Closing (10 minutes)

- Pupils come up with if/else rules to describe the operation of a school lunch system. In the slides, this is specific to the particular school these lessons were trialled in however this activity can be modified to correspond to whatever lunch system is used, even as simple as marking on a register "packed lunch", "school dinner" etc.

Spatial Skills

- Ensure the VPC process is emphasised in the taught section to encourage pupils to visualise the execution of the code, including which branch is selected.
- In the closing activity, encourage pupils to visualise the system of choice. Abstracting the if/else statements from their own visualisation of this process can be seen as a spatial process. Ensure plenty of time (at least 10 minutes) is left to conduct this activity.